Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/04/2007
PI Name:	Bates, Theodore R. Ph.D.		
Project Title:	Oral Absorption and Drug-drug Interaction Kinetics of Quinolone Antibiotics in an Animal Model of Weightlessness		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77004-4501	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: NCE to 5/31/07 (info rec'd 6/1/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Liang, Dong (Texas Southern University)		
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Performance Goal No.:			
Performance Goal Text:			
	The results of previously reported studies conducted in the tail-suspended (TS) rat model indicated that simulated weightless has little effect on the gastrointestinal absorption and disposition kinetics of the water soluble fluoroquinolone antibiotics, ciprofloxacin and moxifloxacin. However, concomitant oral administration of moxifloxacin with Maalox® (a commercial aluminum hydroxide/magnesium hydroxide-containing antacid preparation) markedly reduced the rate and extent of moxifloxacin absorption in both control and TS rats. Quite interestingly, the oral absorption kinetics of moxifloxacin were insignificantly affected by coadministered Pepto-Bismol® (a commercial bismuth subsalicylate-containing antidiarrheal preparation) in TS rats, but were significantly impaired in control rats. The following important questions arose from these pharmacokinetic studies:		

Task Description:	• What is the effect of administration time interval on the in vivo interaction between moxifloxacin and Maalox® or Pepto-Bismol® and, thus, on the absorption kinetics of moxifloxacin under conditions of weightlessness? • What are the effects of weightlessness on the oral absorption kinetics of a poorly soluble, poorly absorbed antibiotic as compared to the water soluble fluoroquinolone antibiotics? • Are the effects of weightless on the disposition kinetics of drugs affected by gender? We have conducted several experiments to address these questions. Specific Aims of this portion of the Project are: (1) To quantitatively assess whether the time interval between moxifloxacin and an aluminum hydroxide/magnesium hydroxide antacid mixture (Maalox®) or between moxifloxacin and the antidiarrheal agent bismuth subsalicylate (Pepto-Bismol®) oral administrations can affect fluoroquinolone antibiotic absorption kinetics in a tail-suspended microgravity rat model. Our hypothesis is that the administration time interval may alter the effect of weightlessness on the extent of the drug-drug interaction between Maalox® or Pepto-Bismol® and quinolone antibiotics. (2) To study the effect of microgravity on the gastrointestinal (GI) absorption kinetics of the model water-insoluble and inherently poorly absorbed drug griseofulvin using a tail-suspended microgravity rat model. Our hypothesis is that weightlessness may have a significant effect on the rate and extent of absorption (bioavailability) of orally administered griseofulvin. (3) To assess whether simulated weightlessness can affect the disposition kinetics of antipyrine, a marker compound for hepatic oxidative metabolism, and whether gender differences exist. The tail-suspended rat model was used to simulate weightlessness. Our hypothesis is that gender differences in gastrointestinal physiology and/or in the hepatic metabolizing enzyme systems may potentially modify drug absorption and/or disposition during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the planned studies will provide mechanistic information on the currently unknown effects of microgravity on gastrointestinal drug-drug interactions and drug absorption, as well as on the currently unknown effect of gender on drug disposition. Data generated from these animal model kinetic studies should provide useful information pertinent to CPR questions in the area of pharmacology.
Task Progress:	1. Co-administration of moxifloxacin and Maalox® to control or TS rats markedly impaired the rate and extent (bioavailability) of moxifloxacin absorption, even when the drug administration was separated by 60 min. However, in TS rats, the degree of this drug-drug interaction was significantly reduced. Co-administration of moxifloxacin with Pepto-Bismol® to control rats markedly impaired the rate but not the extent of moxifloxacin absorption. In striking contrast, there was no significant effect on either the rate or extent of moxifloxacin absorption in TS rats. 2. Simulated weightlessness appreciably and significantly altered several griseofulvin pharmacokinetic parameters. The mean terminal half-lives after oral administration of griseofulvin to control or tail-suspended (TS) rats were significantly longer [4.30 h (P=0.011) for control rats and 6.13 h (P=0.014) for TS rats] than the value observed after intravenous griseofulvin administration. This suggests that griseofulvin exhibits flip-flop kinetics after oral administration. The mean AUC values observed after oral administration of griseofulvin to control and TS rats, dose normalized from 50 mg/kg to 5 mg/kg, were significantly smaller (P<0.02) than the mean value observed after a 5 mg/kg intravenous dose of griseofulvin. This reflects the poor oral absorption characteristics of the drug. The mean Cmax value for griseofulvin in tail-suspended rats was significantly higher than the value observed in control rats. This suggests that simulated weightlessness can appreciably and significantly alter the GI absorption characteristics of a poorly water soluble drug. 3. Simulated weightlessness had statistically significant and marked effects on antipyrine pharmacokinetics after intravenous administration. The mean AUC was increased by 81% [2679 µg•min/mL in control rats vs. 4845 µg•min/mL in TS rats (P < 0.01)]; the mean terminal elimination half-life was prolonged by 74% [75.3 min in control rats vs. 131 min in TS rats (P = 0.01)]; the mean total body clearance was decreased by 45% [7.52 mL/min/kg in control rats vs. 4.16 mL/min/kg in TS rats (P < 0.01)]; and the mean residence time was increased by 58% [110 min in control rats vs. 175 min in TS (P < 0.01)]. No significant difference existed between the two rat groups with respect to the mean apparent volume of distribution [807 ml/kg (control rats) vs. 778 ml/kg (TS rats)]. The results obtained in the present investigation are consistent with the significant decreases in total hepatic cytochrome P-450 content observed in rats aboard both the Cosmos 1887 spaceflight and Skylab 3 space station. The data from antipyrine study suggest that hepatic oxidative enzymatic function is significantly reduced under simulated weightlessness conditions in male rats, but not in female rats.
Bibliography Type:	Description: (Last Updated: 09/04/2007)
Abstracts for Journals and Proceedings	Liang D, Wei B, Ma J, Bates TR. "Antipyrine Pharmacokinetics in a Simulated Weightlessness Rat Model: Gender Differences." Presented at the American Association of Pharmaceutical Scientists Annual Meeting (Abstract #T2312), San Antonio, TX, Oct. 29-Nov. 2, 2006. AAPS Journal. 2006 Oct; 8(S2):Abstract T2312. , Oct-2006
Abstracts for Journals and Proceedings	Wei B, Liang D, Ma J, Bates TR. "Effect of Simulated Weightlessness on the Pharmacokinetics of Griseofulvin After Oral Administration to Rats." Presented at the American Association of Pharmaceutical Scientists Annual Meeting (Abstract #T2302), San Antonio, TX, Oct. 29-Nov. 2, 2006. AAPS Journal. 2006 Oct; 8(S2):Abstract T2302. , Oct-2006

Abstracts for Journals and Proceedings	Wei B, Liang D, Ma J, Bates TR. "Pharmacokinetics of Antipyrine in a Simulated Weightlessness Rat Model." Presented at the American Association of Pharmaceutical Scientists Annual Meeting (Abstract #T2303), San Antonio, TX, Oct. 29-Nov. 2, 2006. AAPS Journal. 2006 Oct; 8,(S2):Abstract T2303. , Oct-2006